WORKED SOLUTION

STEP III 2014, Question 12

Probability and Statistics • Normal distribution

STEP III 2014, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q12

The key idea

Because $y = \mathrm{e}^x$ is strictly increasing, statements about Y turn into statements about X simply by taking logarithms: $Y \le y$ exactly when $X \le \ln y$. That gives the median and the density of Y at once. For the mean, the trick is to complete the square in the exponent so that the integrand becomes another normal density, whose integral we already know is 1.

Solution

Part (i)

Since e^x is strictly increasing, the events $\{Y \le \mathrm{e}^{x_m}\}$ and $\{X \le x_m\}$ are the same event. Hence

$$\mathrm{P}(Y \le \mathrm{e}^{x_m}) = \mathrm{P}(X \le x_m) = \tfrac{1}{2},$$

so the median of Y is

$$y_m = \mathrm{e}^{x_m}.$$

(Medians pass through any increasing function in this way; means do not, as part (iv) shows.)

Part (ii)

Y takes only positive values. For $y > 0$, let G be the cumulative distribution function of Y. Using the same equivalence as before,

$$G(y) = \mathrm{P}(Y \le y) = \mathrm{P}(X \le \ln y) = F(\ln y).$$

Differentiating with the chain rule, and using $F' = f$,

$$g(y) = G'(y) = f(\ln y) \cdot \frac{1}{y} = \frac{f(\ln y)}{y}, \qquad y > 0,$$

as required (and $g(y) = 0$ for $y \leq 0$).

The mode λ is where g takes its maximum. It lies in the open interval $(0, \infty)$, where g is differentiable, so $g'(\lambda) = 0$. By the quotient rule,

$$g'(y) = \frac{f'(\ln y) \cdot \frac{1}{y} \cdot y - f(\ln y)}{y^2} = \frac{f'(\ln y) - f(\ln y)}{y^2}.$$

Since $y^2 \neq 0$, setting $g'(\lambda) = 0$ gives

$$f'(\ln \lambda) = f(\ln \lambda),$$

which is the required condition.

Part (iii)

The function

$$\frac{1}{\sigma\sqrt{2\pi}} \, \mathrm{e}^{-(x-\mu-\sigma^2)^2/(2\sigma^2)}$$

is exactly the probability density function of a $\mathrm{N}(\mu + \sigma^2, \sigma^2)$ random variable: it is the given normal density with μ replaced by $\mu + \sigma^2$. Any probability density function integrates to 1 over the whole real line, which explains the stated integral.

Now

$$\mathrm{E}(Y) = \mathrm{E}(\mathrm{e}^X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^x \, \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} \, \mathrm{d}x.$$

We complete the square in the combined exponent. Expanding $(x - \mu - \sigma^2)^2 = (x - \mu)^2 - 2\sigma^2 x + 2\mu\sigma^2 + \sigma^4$, we get

$$\begin{aligned} x - \frac{(x-\mu)^2}{2\sigma^2} &= -\frac{(x-\mu)^2 - 2\sigma^2 x}{2\sigma^2} \\ &= -\frac{(x-\mu-\sigma^2)^2 - 2\mu\sigma^2 - \sigma^4}{2\sigma^2} \\ &= -\frac{(x-\mu-\sigma^2)^2}{2\sigma^2} + \mu + \tfrac{1}{2}\sigma^2. \end{aligned}$$

The constant $\mathrm{e}^{\mu + \frac{1}{2}\sigma^2}$ comes outside the integral, so

$$\mathrm{E}(Y) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu-\sigma^2)^2/(2\sigma^2)} \, \mathrm{d}x = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2},$$

using the integral explained above. This is the required result.

Part (iv)

We find all three quantities explicitly.

Median. The normal density is symmetric about μ, so $x_m = \mu$, and by part (i) $y_m = \mathrm{e}^{\mu}$.

Mode. Differentiating the normal density,

$$f'(x) = -\frac{x-\mu}{\sigma^2} \, f(x).$$

Ask yourself How do we use the condition from part (ii) when f is never zero?

Since $f(x) > 0$ for every x, we can divide the condition $f'(\ln\lambda) = f(\ln\lambda)$ by $f(\ln\lambda)$ to get

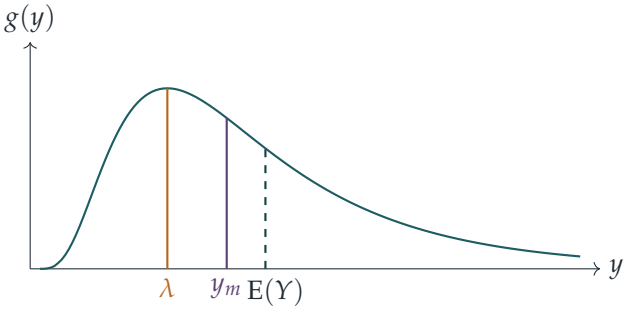
$$-\frac{\ln\lambda - \mu}{\sigma^2} = 1 \quad \Longrightarrow \quad \ln\lambda = \mu - \sigma^2 \quad \Longrightarrow \quad \lambda = \mathrm{e}^{\mu-\sigma^2}.$$

This is the only stationary point of g, and we are told Y has a unique mode, so it is the mode.

Comparison. Since $\sigma > 0$, we have $\mu - \sigma^2 < \mu < \mu + \frac{1}{2}\sigma^2$, and e^t is strictly increasing, so

$$\lambda = \mathrm{e}^{\mu-\sigma^2} \;<\; y_m = \mathrm{e}^{\mu} \;<\; \mathrm{E}(Y) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2},$$

as required. The strict inequalities rely on $\sigma^2 > 0$, which holds for any genuine normal distribution.



The sketch shows the density of Y for $\mu = 0$, $\sigma = 0.6$: the long right-hand tail pulls the mean furthest to the right, the median sits in the middle, and the peak is to the left of both.

Checks

- As $\sigma \to 0$, Y is concentrated at e^{μ} and all three values tend to e^{μ}, as they should.
- With $\mu = 0$, $\sigma = 1$: $\lambda = \mathrm{e}^{-1} \approx 0.368$, $y_m = 1$ and $\mathrm{E}(Y) = \mathrm{e}^{1/2} \approx 1.649$. Numerical integration of $\mathrm{e}^x f(x)$ and a numerical search for the maximum of g agree with these values.
- The ratio $\mathrm{E}(Y)/\lambda = \mathrm{e}^{3\sigma^2/2}$ depends only on σ, which makes sense: changing μ just rescales Y by the factor e^{μ}.

Common slips

- Writing the density of Y as $f(\ln y)$, forgetting the factor $1/y$ from the chain rule. Without it, the mode would come out as e^{μ}, the same as the median.

- Assuming $\mathrm{E}(\mathrm{e}^X) = \mathrm{e}^{\mathrm{E}(X)}$. That is true for the median but not for the mean.
- Sign errors when completing the square: expand $(x - \mu - \sigma^2)^2$ carefully and check the constant left over.

Beyond STEP

Y here has the *lognormal* distribution, which is used to model incomes, share prices and particle sizes. The inequality $\mathrm{E}(\mathrm{e}^X) \geq \mathrm{e}^{\mathrm{E}(X)}$ is a case of Jensen's inequality for convex functions, and the calculation in part (iii) is the moment generating function of the normal distribution, $\mathrm{E}(\mathrm{e}^{tX}) = \mathrm{e}^{\mu t + \frac{1}{2}\sigma^2 t^2}$, evaluated at $t = 1$. Blitzstein and Hwang, *Introduction to Probability*, covers transformations of random variables, moment generating functions and the lognormal distribution.

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